

Steel beam - section A-A

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Span and loading

$$L = 8 \text{ m}$$

$$g_k = \frac{6 \text{ kN}}{\text{m}} = 6 \frac{\text{kN}}{\text{m}}$$

permanent

$$q_k = \frac{9 \text{ kN}}{\text{m}} = 9 \frac{\text{kN}}{\text{m}}$$

imposed

$$w = 1.35 \cdot g_k + 1.5 \cdot q_k = 1.35 \cdot \frac{6 \text{ kN}}{\text{m}} + 1.5 \cdot \frac{9 \text{ kN}}{\text{m}} = 21.6 \frac{\text{kN}}{\text{m}}$$

$$M_{\text{Ed}} = \frac{w \cdot L^2}{8} = \frac{\frac{21.6 \text{ kN}}{\text{m}} \cdot (8 \text{ m})^2}{8} = 172.8 \text{ kN} \cdot \text{m}$$

Material

$$f_y = 355 \text{ MPa}$$

Elastic section modulus of a few rolled sections

PROFILE	H	W _{EL}
IPE300	300 mm	$5.570 \cdot 10^5 \text{ mm}^3$
IPE360	360 mm	$9.040 \cdot 10^5 \text{ mm}^3$
IPE400	400 mm	$1.160 \cdot 10^6 \text{ mm}^3$

Try an IPE360

$$W_{\text{el}} = \text{lookup}(\text{"IPE360"}, \text{sections.profile}, \text{sections.W}_{\text{el}}) = 9.04 \cdot 10^5 \text{ mm}^3$$

$$\sigma = \frac{M_{\text{Ed}}}{W_{\text{el}}} = \frac{172.8 \text{ kN} \cdot \text{m}}{9.04 \cdot 10^5 \text{ mm}^3} = 191.2 \text{ MPa}$$

$$\sigma \leq f_y = 191.2 \text{ MPa} \leq 355 \text{ MPa}$$

OK 46.2% spare

What modulus would just about do?

$$\sigma = f_y \Rightarrow W_{\text{req}} = 4.868 \cdot 10^5 \text{ mm}^3$$

The lightest section that works is the smallest W_{el} above W_{req} .

Wall U-value and heat loss

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Layers, inside out

$$t_{\text{gypsum}} = 13 \text{ mm}$$

$$k_{\text{gypsum}} = \frac{0.25 \text{ W}}{(\text{m} \cdot \text{K})} = 0.25 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

$$t_{\text{ins}} = 195 \text{ mm} \\ \pm 5 \text{ mm}$$

as built, not as drawn

$$k_{\text{ins}} = \frac{0.036 \text{ W}}{(\text{m} \cdot \text{K})} = 0.036 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

$$t_{\text{brick}} = 120 \text{ mm}$$

$$k_{\text{brick}} = \frac{0.77 \text{ W}}{(\text{m} \cdot \text{K})} = 0.77 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

Surface resistances

$$R_{\text{si}} = \frac{0.13 \text{ m}^2 \cdot \text{K}}{\text{W}} = 0.13 \frac{\text{m}^2 \cdot \text{K}}{\text{W}}$$

$$R_{\text{se}} = \frac{0.04 \text{ m}^2 \cdot \text{K}}{\text{W}} = 0.04 \frac{\text{m}^2 \cdot \text{K}}{\text{W}}$$

$$R_{\text{tot}} = R_{\text{si}} + \frac{t_{\text{gypsum}}}{k_{\text{gypsum}}} + \frac{t_{\text{ins}}}{k_{\text{ins}}} + \frac{t_{\text{brick}}}{k_{\text{brick}}} + R_{\text{se}} = \frac{0.13 \text{ m}^2 \cdot \text{K}}{\text{W}} + \frac{13 \text{ mm}}{\frac{0.25 \text{ W}}{\frac{\text{m}}{\text{K}}}} + \frac{195 \text{ mm}}{\frac{0.036 \text{ W}}{\frac{\text{m}}{\text{K}}}} + \frac{120 \text{ mm}}{\frac{0.77 \text{ W}}{\frac{\text{m}}{\text{K}}}} + \frac{0.04 \text{ m}^2 \cdot \text{K}}{\text{W}} \\ \pm 0.1389 \text{ m}^2 \cdot \text{K/W}$$

$$U = \frac{1}{R_{\text{tot}}} = \frac{1}{\frac{5.795 \text{ m}^2 \cdot \text{K}}{\text{W}}} = 0.1726 \frac{\text{W}}{\text{m}^2 \cdot \text{K}} \\ \pm 0.004137 \text{ W/m}^2/\text{K}$$

$$U \leq \frac{0.18 \text{ W}}{(\text{m}^2 \cdot \text{K})} = 0.1726 \frac{\text{W}}{\text{m}^2 \cdot \text{K}} \leq 0.18 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

OK 4.1% spare

Heat through one facade

$$A = 42 \text{ m}^2$$

$$\text{dT} = 22 \text{ K}$$

a temperature difference, so K and not degC

$$Q_{\text{loss}} = U \cdot A \cdot \text{dT} = \frac{0.1726 \text{ W}}{\frac{\text{m}^2}{\text{K}}} \cdot 42 \text{ m}^2 \cdot 22 \text{ K} = 159.5 \text{ W} \\ \pm 3.822 \text{ W}$$

How thin could the insulation get before the limit is missed?

$$U = \frac{0.18 \text{ W}}{(\text{m}^2 \cdot \text{K})} \Rightarrow t_{\min} = 186.4 \text{ mm}$$